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A Latent Space Graded Response Model for Likert-Scale Psychological Assessments

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ABSTRACT

In this study, we introduce a novel modeling approach for ordinal response data, extending the one-parameter graded response model. The proposed model incorporates unobserved interactions between respondents and items, represented as distances in a two-dimensional Euclidean space, referred to as an interaction map. This latent space graded response model (LSGRM) addresses potential violations of the conditional independence assumption shared by traditional main-effect-only psychometric models and offers a visualization tool for exploring conditional dependence in ordinal item response data. Through simulation and empirical studies, we illustrate the utility of the proposed approach in analyzing Likert-scale psychological assessment data. Also, by comparing the results with those from other models of different data modalities, we examined the impact of dichotomization and treating ordinal responses as continuous on conditional dependence.

KEYWORDS

Likert scales; psychological assessment; conditional dependence; latent space; graded response model

Introduction

Likert scales in psychological assessment

Likert scales are one of the most prevalent tools in psychological assessments, enabling the measurement of attitudes, opinions, and various psychological constructs. These scales typically consist of a series of statements, where respondents indicate their level of agreement or frequency of occurrence on a defined scale, often ranging from *strongly disagree* to *strongly agree* or from *never* to *always* (Carifio & Perla, 2007; Jebb et al., 2021). The structured nature of Likert scales allows for quantifying subjective experiences, which is particularly useful in clinical and research settings. This quantification is crucial for assessing symptoms, treatment outcomes, and the overall impact of psychological conditions on daily functioning (Jebb et al., 2021).

The versatility of Likert scales allows them to be adapted for various psychological constructs, such as self-esteem, anxiety, and well-being, and so be applied across different domains of psychological research (Qadeer et al., 2023; Zhang et al., 2019). For instance, in personality psychology, various personality traits

have been measured by carefully developed item inventories based on the Likert scale, such as the Big-five inventory (Goldberg, 1992). Also in clinical psychology, the Likert scale is frequently employed to assess the severity of psychological conditions, track treatment outcomes, or measure the intensity of symptoms (Preston & Colman, 2000).

Conditional dependence under Likert scales and their psychometric models

Traditional models for Likert scale data, such as the Graded Response Model (GRM; Samejima, 1969), typically assume that variability in responses is driven solely by the main effects of persons and items. This assumption, known as conditional independence (CI), is a cornerstone of most traditional psychometric models. It posits that item responses are independent of one another when conditioned on the latent traits and item parameters. In other words, the assumption implies that all associations among item responses can be perfectly explained by person-specific latent variables and item-specific parameters.

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However, this assumption may fail to hold in practice. For instance, certain symptoms from clinical assessments may be more closely related than others, even after accounting for the overall level of endorsement. Moreover, the CI assumption implies that individuals with identical latent trait levels are equally likely to endorse specific items. This overlooks the possibility that some individuals might have a greater tendency to endorse particular symptoms than others, even when their overall trait levels are similar. If such differences in endorsement tendency are suspected across groups, some previous approaches, such as tests for differential item functioning (Magis et al., 2010) and measurement (in)variance (Meredith, 1993; Molenaar, 2021), can be employed. However, violations of the CI assumption, namely the occurrence of conditional dependence (CD), are a broader concept encompassing not only across-group differences in measurement but also across-person differences in how items behave.

Consequently, it is important to examine why CD occurs and what it implies for psychological and educational assessment. When substantial CD is present that cannot be accounted for by parameters in traditional psychometric models, respondents may perceive item symptoms and behaviors differently than described by the item parameters. Also, person-specific traits may vary across items. These unexplained variations, therefore, can provide detailed information for heterogeneous item properties and tendencies of respondents, which can be utilized for personalized diagnoses, evaluations, and feedback (Jeon et al., 2021; Kang & Jeon, 2025b). In this sense, understanding the underlying causes of CD and finding ways to effectively account for or model these dependencies becomes crucial.

Some sources of CD can be anticipated from ordinary psychological and educational measurements. For instance, CD can arise from the nature of the assessment itself. Respondents may perceive the thresholds between adjacent response categories inconsistently across items, leading to unexpected dependencies. More specifically, consider a job satisfaction survey where respondents rate their agreement with statements such as *I am satisfied with the recognition I receive for my work* (Item A) and *I am satisfied with my chances for promotion* (Item B) on a Likert scale from 1 (*Strongly Disagree*) to 5 (*Strongly Agree*). The ordinal nature of the scale may lead respondents to interpret the transition between categories (e.g., from *Agree* to *Strongly Agree*) differently for each item. A respondent might perceive this shift on Item A as a

significant increase in satisfaction, while viewing the same shift on Item B as a minor adjustment. This non-linear interpretation introduces CD, as the correlations between items are influenced not only by the underlying latent trait (e.g., job satisfaction) but also by individual differences in how response categories are perceived and utilized.

CD can also arise from item wording in Likert scale assessments, especially when items emphasize similar behaviors or characteristics, leading to a dependence between responses not fully captured by the latent trait. For example, in the Five Factor personality assessment (Goldberg, 1992; Schmit & Ryan, 1993), which measures the trait of *Conscientiousness*, items like *I am detail-oriented* and *I am thorough in my work* are intended to measure the same underlying trait. However, the wording of these items may seem to some respondents excessively similar, beyond what can be expected from their common latent trait. This similarity can create shared variance that is not solely attributable to the latent trait of *Conscientiousness*. Instead, the observed correlation between these responses reflects the overlap in their meaning rather than variability in the underlying trait.

Model misspecification and conditional dependence

Along with the sources of CD described above, CD can arise due to a bit unnatural but rather frequently observed reason in psychometric modeling: model misspecification. There could be various types of such inaccurate representations of latent structure underlying item responses, but we particularly focus on misspecifications of the link function used in ordinal data analysis. For ordinal data analysis, appropriate choices for the link function include multinomial logit, probit, etc. However, sometimes ordinal responses from the Likert scales have been analyzed by 1) applying inappropriate dichotomization (e.g., a median split) and a binary logit link function, and 2) treating them as continuous or interval and then using an identity link function. These misspecifications lead to biased estimates of the main person and item effect parameters, potentially leading to artificial dependencies between items, violations of the CI assumptions, and misjudgments of the degree and configurations of CD.

Dichotomizing Likert scale data

Dichotomization involves transforming ordinal responses into binary categories by setting arbitrary thresholds, such as the observed median (Iacobucci et al., 2015) and the

midpoint of the scale (Arends-Tóth & van de Vijver, 2006), or, by excluding neutral responses to highlight polarized opinions (top/bottom box approach; Li et al., 2010). Regardless of which approach is employed, dichotomization can diminish the richness of the data, which can lead to improper statistical results such as diminished effect size and power, occurrences of spurious statistical significance, overlooking of potential nonlinear associations between variables, severe declines in measurement reliability, and artificially reduced individual differences in latent variable models. (MacCallum et al., 2002; Maxwell & Delaney, 1993; McClelland et al., 2015; Royston et al., 2006; Rucker et al., 2015; Schweizer et al., 2021).

Dichotomization can also impact CD of psychometric models when applied to ordinal item responses. This process fundamentally alters variable types and the underlying data structure, which can lead to substantial differences in the information that can be extracted from the dataset. Moreover, after dichotomization, an item response theory (IRT) model such as the Rasch model must be used instead of the GRM designed for ordinal responses. Although these two models share some similarities, they would produce heterogeneous estimates of latent traits and item parameters, potentially resulting in different conclusions regarding the existence of CD and its implications.

Treating Likert scales as continuous

Treating Likert scales as continuous leads to incorrect assumptions about the underlying distribution (e.g., from ordered multinomial to normal). Consequently, this can distort the results of statistical analyses, including inflated Type I and II errors, reduced effect sizes, and even *inversions* of true ordering of effects (Liddell & Kruschke, 2018; Russell & Bobko, 1992). It has also been known that computing Pearson correlations by treating ordinal responses as continuous can yield results that differ substantially from those obtained using polychoric correlations (Olsson, 1979).

Likewise, applying continuous-response models to Likert scale data can severely distort CD. For example, when a factor analysis model is used instead of the GRM or other psychometric models for ordinal responses, main effect parameters such as latent variables and factor loadings cannot properly represent the true person and item characteristics. This can obscure the extent of uncaptured information in data (i.e., CD), alter its statistical properties, and disrupt its substantive implications. In this regard, studying CD underlying Likert-based assessment data requires a psychometric model that more appropriately represents the response types.

A new approach

In this article, we propose a new modeling approach for Likert scale assessment, the Latent Space Graded Response Model (LSGRM). The most important feature of the proposed model is that it is designed to capture CD when the data violates the CI assumption of the traditional GRM. While estimating the main effects (latent variables and item parameters) in the GRM, the LSGRM also captures unobserved interactions between respondents and items by embedding them in a two-dimensional metric space, referred to as an interaction map. This map visualizes the underlying structure of CD in the data by placing the latent positions of respondents and items within a shared latent space. The estimated positions can be used to compute respondent-item distances, which lead to the quantification of their interactions.

The proposed LSGRM is an integration of the traditional GRM with a latent space approach originally proposed for network data analysis (Hoff et al., 2002), but adopted to account for CD and respondent-item interactions in the form of their measurement network. It is worth noting that the same idea of combining latent space into traditional psychometric models has been introduced in several previous studies, such as the latent space item response model (LSIRM; Jeon et al., 2021) for binary responses and the latent space confirmatory factor analysis (LSCFA; Kang & Jeon, 2025a) model for continuous responses. Thus, our proposed model extends 1) the same stream of research developing new psychometric models for respondent-item interactions, but is appropriate for Likert scales and 2) previous investigations on the effect of model misspecifications for Likert scales (treating them as continuous or dichotomizing them), but in the context of CD, which to our knowledge has not been widely studied in the field.

The paper is structured as follows. We first formally introduce the proposed model, with its structure and estimation method. A series of simulation studies follows, which aims to validate parameter recovery of the LSGRM and examine the accuracy of the model in detecting CD when it substantially exists under ordinal response data and rejecting CD when it does not. Then, we proceed with two empirical applications of the proposed model to ordinal response data. The first application to Maternal Depression data illustrates the practical advantage and utility of the LSGRM in capturing the complex CD in the form of interactions between symptoms and respondents in a real-world dataset. In the second application to Internet Addiction Test data, we compare the LSGRM

with the LSIRM fitted to dichotomized data and the LSCFA model fitted to the original ordinal response data but treating them as continuous, highlighting the differences in the results (focusing on CD) due to model misspecification. We conclude with a summary of the main findings and a discussion on possible avenues for future research.

Model

Graded response model

The GRM (Samejima, 1969) is a widely used item response theory (IRT) model for ordinal responses. A one-parameter GRM can be written as follows:

$$\text{logit } P(Y_{ji} \geq k) = \theta_j - \beta_{ik} \quad (1)$$

where:

- Y_{ji} is the ordinal response of respondent j ($j = 1, \dots, N$) to item i ($i = 1, \dots, I$) which can take one of $k = 1, \dots, K$ for the K -point Likert scale.
- $P(Y_{ji} \geq k)$ is the probability that respondent j responds in category k or higher to item i , given that $P(Y_{ji} = k) = P(Y_{ji} \geq k) - P(Y_{ji} \geq k + 1)$, with $P(Y_{ji} \geq 1) = 1$ and $P(Y_{ji} > K) = 0$.
- $\theta_j \in \mathbb{R}$ is the latent trait (e.g., depression severity) of respondent j and $\theta_j \sim N(0, \sigma^2)$.
- $\beta_{ik} \in \mathbb{R}$ ($k = 1, \dots, K - 1$) is the threshold to decide between categories k and $k + 1$ for item i .

Note that the one-parameter GRM in Equation 1 excludes item discrimination parameters to maintain consistency with the formulation of the proposed modeling approach, which is detailed in the following section. The GRM assumes that ordinal responses are conditionally independent of one another, given the person parameters (i.e., latent traits) and item parameters (i.e., item thresholds). However, as previously discussed, this assumption can be violated in real-world applications, leading to biased inferences about both individuals and items (Kang & Jeon, 2025b).

Latent space graded response model

Model definition

The proposed LSGRM is an extension of the one-parameter GRM (Equation 1) to address violations of the strict CI assumption. The extended model, along with the latent trait θ_j and the item threshold β_{ik} in GRM, incorporates interactions between persons and items in the form of their distances in a latent space.

The proposed approach can also be seen as an extension of the LSIRM (for binary responses) and the LSCFA model (for continuous responses) to ordinal responses.

The first primary assumption of the LSGRM is that both respondents and items are embedded in a shared space (i.e., a latent space). In this article, we assume a two-dimensional real-numbered latent space $\mathbb{R}^2$, as done in previous studies both for network data analysis (e.g., Handcock et al., 2007; Hoff et al., 2002; Smith et al., 2019) and for psychometric modeling (e.g., Jeon et al., 2021; Kang & Jeon, 2025a). Respondents and items have their positions on the latent space, which we denote as $\mathbf{z}_j$ (for person j) and $\mathbf{w}_i$ (for item i), respectively.

The second primary assumption of the LSGRM is that the distance between person j and item i on the latent space, denoted as $d(\mathbf{z}_j, \mathbf{w}_i)$, represents person-item interactions unobserved by latent trait θ_j and item threshold β_{ik} . Specifically, a larger distance indicates that item i does not well describe the trait of person j , leading to lower responses on the Likert scale, beyond what is predicted by θ_j and β_{ik} (i.e., prediction from the GRM). Distance of zero indicates that there is no such interaction between person j and item i and so the person j 's response to item i can be well described by the main person and item effects parameters (θ_j and β_{ik}).

Taken together, the proposed LSGRM can be expressed as follows:

$$\begin{aligned} \text{logit } P(Y_{ji} \geq k) &= \theta_j - \beta_{ik} + f(\mathbf{z}_j, \mathbf{w}_i) \\ f(\mathbf{z}_j, \mathbf{w}_i) &= -\gamma \cdot d(\mathbf{z}_j, \mathbf{w}_i) \end{aligned} \quad (2)$$

where:

- Y_{ji} , θ_j , and β_{ik} are defined in the same way as done for Equation 1.
- $\mathbf{z}_j \in \mathbb{R}^2$ and $\mathbf{w}_i \in \mathbb{R}^2$ represent the latent positions of respondent j and item i in a latent space, respectively.
- $d(\cdot, \cdot) \geq 0$ is a distance function, assigning non-negative distances to pairs of points $(\mathbf{z}, \mathbf{w})$ in the latent space. Specifically, we consider the Euclidean distance $d(\mathbf{z}_j, \mathbf{w}_i) = \|\mathbf{z}_j - \mathbf{w}_i\|_2 = \sqrt{\sum_{l=1}^2 (z_{jl} - w_{il})^2}$, which satisfies the following properties:
 - Reflexivity: $d(\mathbf{z}, \mathbf{w}) = 0$ if and only if $\mathbf{z} = \mathbf{w} \in \mathbb{R}^2$.
 - Symmetry: $d(\mathbf{z}, \mathbf{w}) = d(\mathbf{w}, \mathbf{z})$ for all $\mathbf{z}, \mathbf{w} \in \mathbb{R}^2$.
 - Triangle inequality: $d(\mathbf{z}, \mathbf{w}) \leq d(\mathbf{z}, \mathbf{v}) + d(\mathbf{w}, \mathbf{v})$ for all $\mathbf{z}, \mathbf{w}, \mathbf{v} \in \mathbb{R}^2$.

- $\gamma \geq 0$ is a weight/tuning parameter of the distance term.
 - With a preceding negative sign, $\gamma > 0$ ensures that the distance term functions as a penalty term, such that a larger distance between person j and item i is associated with decreases in logit $P(Y_{ji} \geq k)$, the logit of the probability of person j endorsing response category k or higher for item i .
 - This term also works as a scaling term, which accounts for the scale difference between the logit probability and the distance in the latent space
 - Notably, when $\gamma = 0$, the model reduces to the one-parameter GRM in Equation 1

The interpretation of the interaction term $\gamma \cdot d(\mathbf{z}_j, \mathbf{w}_i)$ is central to the utility of the proposed model. As noted earlier, a greater distance between $\mathbf{z}_j$ (the latent position of person j) and $\mathbf{w}_i$ (the latent position of item i) decreases the probability of the respondent endorsing category k or above for the item i , given their overall latent trait θ_j level and the item thresholds β_{ik} . This implies that even the respondents j and j' with very similar latent trait levels $\theta_j \approx \theta_{j'}$ can exhibit largely different response probabilities, depending on their relative positions to items in an interaction map. Similarly, items with comparable threshold levels may have varying probabilities of category k or above being selected due to their distances to respondents in an interaction map. In essence, the model captures the *interactions* between respondents and items that cannot be captured by θ_j and β_{ik} , by means of the latent space and the distance effect, thus relaxing the CI assumption inherent in traditional, main-effect-only models.

The latent space should not be confused with the factor space (i.e., the space of latent traits), as it is a space of interactions and not directly related to θ_j . The person latent position $\mathbf{z}_j$ is distinct from the latent trait θ_j . While θ_j captures overall trait levels of person j , $\mathbf{z}_j$ (with its distances to $\mathbf{w}_i, i = 1, \dots, I$) accounts for person-item interactions uncaptured by θ_j . Similarly, the item latent position $\mathbf{w}_i$ is distinct from the item thresholds β_{ik} . While β_{ik} captures the overall item effect of item i , $\mathbf{w}_i$ is introduced to account for person-item interactions. Relatedly, it is worth noting that it would be less meaningful to interpret $\mathbf{z}_j$ independently, without understanding its interactions with $\mathbf{w}_i$. This is because 1) what matters in the proposed model is the distance $d(\mathbf{z}_j, \mathbf{w}_i)$ as it captures person-item interactions and 2) the term $\mathbf{z}_j$ does not work in isolation as it influences the logit response probability only through the distance term

$d(\mathbf{z}_j, \mathbf{w}_i)$. Instead, $\mathbf{z}_j$, as well as $\mathbf{w}_i$, can be seen as a statistical tool to capture unobserved interactions underlying CD.

Furthermore, although the model explicitly focuses on modeling person-item distances and person-item interactions, the triangle inequality of latent positions allows for the interpretation of distances between items as well as those between respondents within the interaction map. Specifically, these distances reflect unobserved similarities and dissimilarities. For instance, if items A and B are positioned closely together in the interaction map, while item A is far from item C, it indicates that items A and B are more similar to each other than to item C. These similarities may not be fully explained by their threshold levels, highlighting additional insights provided by the proposed model with an interaction term.

Related models

Full interaction model

The proposed model is a special case of the GRM with an interaction term, namely logit $P(Y_{ji} \geq k) = \theta_j - \beta_{ik} + \eta_{ji}$ where η_{ji} is an interaction term for person j and item i . Estimating this full interaction model is not feasible, because it requires finding optimal values for $(N \times I)$ of η_{jk} with a $(N \times I)$ response matrix. The LSGRM resolves this issue by imposing a constraint on the interaction with the distance effect: $\eta_{ji} = -\gamma \cdot d(\mathbf{z}_j, \mathbf{w}_i)$. In this sense, the LSGRM captures and estimates person-item interactions in a structured manner. Also, the model is constrained in that it cannot fully capture all interactions if some response variations due to interactions underlying CD cannot be well represented by the distance effect.

Other latent space measurement models

As mentioned above, there are several latent space measurement models grounded in the same principles as the proposed LSGRM. The LSIRM (Jeon et al., 2021; see also Kang & Jeon, 2024 and Luo et al., 2023) was the first model accounting for CD underlying binary response data by integrating the latent space into the standard Rasch model. This approach allows for the visualization and analysis of respondent-by-item interactions that the Rasch models do not fully explain.

Other latent space models have extended this approach within different psychometric frameworks. For example, the Latent Space Diffusion Item Response Theory (LSDIRT) model, introduced by

Kang et al. (2023) accounts for respondent-item interactions underlying response accuracy and response times (RTs) by embedding both respondents and items within a latent space. This model expands on the Diffusion Item Response Theory (DIRT; Kang et al., 2022; Tuerlinckx & De Boeck, 2005; van der Maas et al., 2011) model with the latent distances to modulate the information processing rate (so-called drift rate) in a cognitive process of a person responding to an item, thus capturing interactions that influence both response accuracy and RTs through variations of efficiency in information processing. By combining a process model and the latent space approach, the LSDIRT model offers a comprehensive joint modeling approach that accounts for variations in both responses and RTs and provides a theoretical representation of the cognitive processes of a respondent in psychological measurement.

Similarly, the LSCFA (Kang & Jeon, 2025a) model extends latent space modeling to non-binary responses by combining it with the Confirmatory Factor Analysis (CFA) model. While the LSIRM focuses on the presence of a condition or behavior, the LSCFA model generalizes the approach to handle broader psychometric assessments. Through distances between persons and items, the LSCFA captures person-item interactions unexplained by factor space, factor loadings, and factor correlations, allowing researchers to explore residual dependencies that factor models alone may miss.

Each of these latent space models, LSIRM, LSDIRT, and LSCFA, offers unique contributions to psychometrics by addressing different types of data and modeling requirements, all while capturing person-item interactions in ways that traditional models cannot. These approaches provide deeper insights into how respondents perceive items and the resulting information can be further utilized to produce customized diagnoses, evaluations, and feedback for respondents as well as items. The proposed LSGRM can extend these contributions by addressing the Likert scale assessment and potential CD that can underlie ordinal responses.

Bayesian estimation

We estimate the LSGRM using a fully Bayesian approach. In the current article, we used the Hamiltonian Monte Carlo (HMC) method implemented in **Stan** (Stan Development Team, 2025). While users can specify their own prior distributions given preferences or information from previous relevant studies, we recommend the following specifications of non- or weakly-informative priors in case there is no such information:

- $\theta_j \mid \sigma^2 \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \sigma^2 > 0, j = 1, \dots, N.$
- $\beta_{ik} \mid \tau_{\beta}^2 \stackrel{\text{iid}}{\sim} MVN_k(\mathbf{0}, \tau_{\beta}^2), \tau_{\beta}^2 > 0, i = 1, \dots, I, k = 1, \dots, K - 1.$
- $\log(\gamma) \mid \tau_{\gamma}^2 \sim N(\mu_{\gamma}, \tau_{\gamma}^2), \mu_{\gamma} \in \mathbb{R}, \tau_{\gamma}^2 > 0.$
- $\sigma^2 \mid a_{\sigma}, b_{\sigma} \sim \text{Inv-Gamma}(a_{\sigma}, b_{\sigma}), a_{\sigma} > 0, b_{\sigma} > 0.$
- $\mathbf{z}_j \stackrel{\text{iid}}{\sim} MVN_p(\mathbf{0}, \mathbf{I}_p), j = 1, \dots, N, p = 2.$
- $\mathbf{w}_i \stackrel{\text{iid}}{\sim} MVN_p(\mathbf{0}, \mathbf{I}_p), i = 1, \dots, I, p = 2.$

where $\mathbf{0}$ represents a zero vector with an appropriate dimension and $\mathbf{I}_p$ is the $p \times p$ identity matrix. The hyperparameters are chosen as follows: $\tau_{\beta}^2 = 10, a_{\sigma} = 1, b_{\sigma} = 1, \mu_{\gamma} = 0.5, \tau_{\gamma}^2 = 1$. While we employ **Stan** in the current study, other Bayesian programs can also be utilized to estimate the latent space models adapted for psychological assessment data. A comparative study of such programs can be found in Luo et al. (2023).

Identifiability

In latent space modeling, distances are invariant to translations, reflections, and rotations of the latent positions of respondents and items, leading to an identifiability issue. To resolve this, we apply Procrustes matching as a post-processing step for the posterior samples of the latent positions, by choosing the posterior sample of latent positions with the highest log posterior density as a reference set, following the approach of previous studies (Jeon et al., 2021; Jin & Jeon, 2019). Subsequently, all other posterior samples are aligned to this reference set.

In addition, it is a common practice to assign priors to the latent positions $\mathbf{z}_j$ and $\mathbf{w}_i$ rather than directly to the distances in latent space modeling. Specifying a prior for distances is more complex because they must adhere to the triangle inequality. We set the prior mean of the latent positions to 0 to fix the center of the latent space. Further, to estimate the weight parameter γ , which in essence determines the scale of the latent space, we fix the prior variance of the latent positions to 1.

Spike-and-slab prior for model selection

The weight parameter γ quantifies the influence of the distance term, i.e., person-by-item interactions, on the response probability. When $\gamma = 0$, the LSGRM simplifies to the traditional one-parameter GRM. As such, determining whether $\gamma = 0$ for the data being analyzed can be of interest. If $\gamma = 0$, it indicates minimal person-by-item interaction within the data, suggesting that the traditional GRM is sufficient. Conversely, if

$\gamma > 0$, it implies the presence of non-negligible person-by-item interactions, highlighting the value of exploring these interactions to better understand the structure and nature of CD in the data.

Since $\gamma = 0$ reduces the LSGRM to the traditional one-parameter GRM, determining whether $\gamma = 0$ or not essentially becomes a model selection problem between the LSGRM and the GRM. To address this, we adopt a spike-and-slab approach (Ishwaran & Rao, 2005), assigning the following prior to the log-transformed tuning parameter $\log(\gamma)$:

$$\log(\gamma) \sim (1 - \delta)N_{\text{spike}}(\mu_{\gamma_0}, \tau_{\gamma_0}^2) + \delta N_{\text{slab}}(\mu_{\gamma_1}, \tau_{\gamma_1}^2),$$

$$\delta \in \{0, 1\}$$
(3)

If no substantial CD is found in the data, the spike part shrinks γ to 0, removing the interaction term and reducing the model to the GRM. On the other hand, if the substantial CD is found, the slab part is selected, allowing γ to take positive values and the distance terms to naturally capture respondent-item interactions.

The choice of the spike or slab part in Formula 3 depends on δ , which is simultaneously estimated with all the other model parameters. Respondent-item interactions can be considered meaningful if the posterior inclusion probability (PIP) $P(\delta = 1 | \gamma)$ is large. If PIP is less than a cutoff value (e.g., 0.5), the GRM can be chosen. Otherwise, the LSGRM can be selected and interactions can be further examined with estimated latent positions and their distances.

One complication in using **Stan** to estimate the model is that the program does not allow sampling discrete parameters such as δ in the spike-and-slab prior. Instead, δ must be marginalized out with a hyperprior imposed on the prior inclusion probability $\omega = P(\delta = 1)$. Then, the joint posterior distribution including ω can be estimated using MCMC methods. Also, the PIP can be computed from the Bayesian chains of γ and ω as:

$$P(\delta = 1 | \gamma, \omega, \mu_{\gamma_0}, \tau_{\gamma_0}^2, \mu_{\gamma_1}, \tau_{\gamma_1}^2)$$

$$= \frac{\omega \cdot N(\gamma | \mu_{\gamma_1}, \tau_{\gamma_1}^2)}{(1 - \omega) \cdot N(\gamma | \mu_{\gamma_0}, \tau_{\gamma_0}^2) + \omega \cdot N(\gamma | \mu_{\gamma_1}, \tau_{\gamma_1}^2)}$$
(4)

where $N(\cdot | \mu, \tau^2)$ indicates a probability density function of a normal distribution with mean μ and variance τ^2 .

Simulation studies

We present two simulation studies: one designed to validate the Bayesian estimation of the proposed model and the other to assess the effectiveness of the model selection method using the spike-and-slab prior.

Study 1: Validation of latent space graded response model

The first simulation aims at evaluating the parameter recovery of the proposed model. To this end, we consider the following simulation conditions:

- Number of respondents: $N = 300, 500, 1000$.
- Number of items: $I = 15, 25$.
- Number of response level: $K = 4, 5, 6, 7$.

Combinations of these factors result in a total of $3 \times 2 \times 4 = 24$ simulation conditions. For each of the conditions, data-generating parameter values are sampled or determined as follows:

- Latent trait: $\theta_j \sim N(0, \sigma_\theta^2)$, $j = 1, \dots, N$, with $\sigma_\theta = 1.25$.
- Item thresholds: $\beta_{ik} \sim N(\mu_\beta, \tau_\beta^2)$, $i = 1, \dots, I$ and $k = 1, \dots, K - 1$, with $\tau_\beta^2 = 0.5$ and μ_β of
 - $K = 4$: $(-5, -3, -1.5)$
 - $K = 5$: $(-5, -3.5, -1.5, 0)$
 - $K = 6$: $(-5, -3, -2, -1, 0)$
 - $K = 7$: $(-5, -4, -3, -2, -1, 0)$
 and the order constraint of $\beta_{i1} < \beta_{i2} < \dots < \beta_{i,K-1}$.
- Latent positions: $\mathbf{z}_j \sim MVN_2(\mathbf{0}, \mathbf{I}_2)$, $j = 1, \dots, N$, and $\mathbf{w}_i \sim MVN_2(\mathbf{0}, \mathbf{I}_2)$, $i = 1, \dots, I$.
- Distance weight parameter: $\gamma = 1.4$.

These distributions and values are determined by the parameter estimates that we obtained from the LSGRM fit to our empirical datasets, which will be introduced in Section 4. Specifically, we have taken values between the parameter estimates from two example datasets (e.g., values roughly close to their averages) so that the resulting data-generating parameter values can produce synthetic datasets that resemble real-world item responses. With the obtained data-generating parameter values, 50 sets of ordinal responses are generated from the LSGRM per simulation condition.

Then, we fit the LSGRM to each of the simulated datasets, using the Bayesian estimation method described in Section 2.4. We run three Bayesian chains for 5000 iterations with the first 2500 iterations discarded for burn-in, resulting in 7500 posterior samples per repetition. We evaluate the convergence of the MCMC chains with traceplots and the potential scale reduction factor ($\hat{R}$; Gelman, 1996; Gelman et al., 2013) with the cutoff of 1.1, supporting satisfactory convergence.

To evaluate parameter recovery, we calculate point estimates of parameters using the posterior means for each repetition. Then, we compute the absolute bias, standard error (SE), and mean squared error (MSE) of each parameter. To summarize the results of person and items parameters, we average estimated values across persons (for θ_j and z_j) and across items (for β_{ik} and w_i). Additionally, estimated values for item thresholds are averaged across $K - 1$ category levels and those for latent positions are averaged across $p = 2$ dimensions. Also, for each repetition, we compute correlations between the true and estimated parameter values, across persons for person parameters and across items for item parameters. Then, the mean and standard deviation (SD) of the correlations across 50 repetitions are obtained.

Table 1 summarizes the parameter recovery results when $K = 4$ (results for $K = 5, 6$, and 7 can be found in Section A in our [Supplementary Materials](#)). The

Table 1. Parameter recovery results when response level is $K = 4$.

Indices	N	I	θ_j	β_{ik}	σ	z_j	w_i	γ
MSE	300	15	0.443	0.180	0.004	0.408	0.090	0.014
		25	0.362	0.226	0.003	0.310	0.098	0.014
	500	15	0.424	0.130	0.003	0.396	0.063	0.012
		25	0.281	0.132	0.004	0.237	0.068	0.009
	1000	15	0.471	0.104	0.001	0.428	0.042	0.004
		25	0.359	0.080	0.002	0.294	0.034	0.002
Bias	300	15	0.403	0.238	0.032	0.322	0.131	0.095
		25	0.324	0.269	0.033	0.239	0.132	0.103
	500	15	0.369	0.165	0.030	0.312	0.089	0.102
		25	0.271	0.141	0.053	0.183	0.125	0.086
	1000	15	0.431	0.132	0.019	0.330	0.068	0.033
		25	0.347	0.091	0.043	0.235	0.064	0.027
SE	300	15	0.417	0.320	0.057	0.452	0.231	0.072
		25	0.398	0.313	0.046	0.414	0.203	0.059
	500	15	0.421	0.287	0.041	0.447	0.197	0.046
		25	0.384	0.273	0.029	0.387	0.176	0.045
	1000	15	0.400	0.211	0.030	0.446	0.154	0.059
		25	0.380	0.198	0.022	0.408	0.131	0.040
Corr	300	15	0.840	0.932		0.775	0.972	
		25	0.881	0.909		0.836	0.970	
	500	15	0.851	0.934		0.789	0.981	
		25	0.900	0.933		0.872	0.983	
	1000	15	0.829	0.940		0.758	0.986	
		25	0.870	0.960		0.844	0.988	

MSE, Absolute Bias, and SE were calculated for each parameter and then averaged over persons, items, and/or parameter dimensions. The correlations between true and estimated parameter values averaged across repetitions were also shown.

results indicate reasonable degrees of consistency between the true and estimated values of the model parameters. Indices also exhibit the typically anticipated effects of simulation conditions. For instance, the person parameters are better recovered with the larger number of items I , and the item parameters are better recovered with the larger number of persons N . MSEs and Biases of the item thresholds are generally comparable or slightly increased compared to those presented in the recovery simulation of the GRM (i.e., when the data-generating model was the GRM with the CI assumption, not the LSGRM with CD; Kieftenbeld & Natesan, 2012; Kuo & Sheng, 2016; Reise & Yu, 1990). Summary statistics of person parameters are increased more. For instance, Kieftenbeld and Natesan (2012) reported that the average absolute bias for the latent trait when $I = 15$ was 0.260 (compared to about 0.401 under the same I condition, averaged across the three N conditions for the LSGRM). This can be attributed to the increased model complexity of the proposed model, as it incorporates more parameters, such as latent positions.

In the comparison between the LSGRM recovery results and the previously reported GRM results, the recovery results of the LSGRM were under the substantial CD, whereas those of the GRM were under the CI assumption. Therefore, directly comparing those results may not provide a clearer view of the model recovery performance. To address this issue, we extended our investigation with cross recovery. Specifically, we fitted the GRM to the synthetic data generated from the LSGRM (those used to generate Table 1). Also, we generated other sets of synthetic data from the GRM with the CI assumption, and fitted both the GRM and the LSGRM to them. As a result, this cross recovery study can directly compare the recovery performance of the models 1) under the CD condition (when there is substantial CD) and 2) under the CI condition (when the CI assumption is met). The table results of the cross recovery are provided in our [Supplementary Materials](#) (Section A), with visualized results presented below.

Figure 1 presents the cross recovery results under the CD condition. The left, middle, and right panels show the trend of MSE, Bias, and SE (as denoted on the top of the panels), respectively, as a function of sample size conditions (P and I , as denoted on the bottom of the panels). In each panel, the red triangles with dot-dashed lines connecting them represent the results from the GRM whereas the blue dots with solid connecting lines indicate those from the

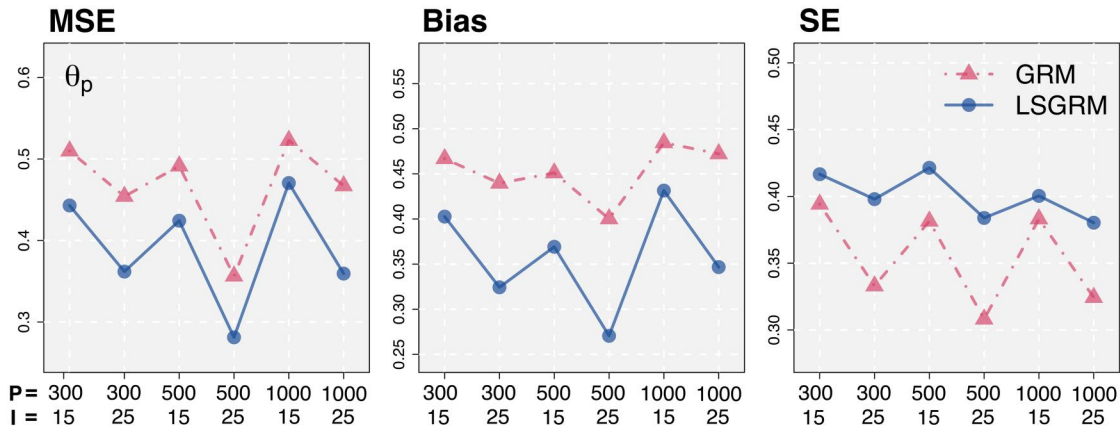


Figure 1. Cross recovery results of the proposed LSGRM and the traditional GRM under the CD condition (i.e., when the data-generating model was the LSGRM).

LSGRM, as indicated by the legend in the rightmost panel.

It is noteworthy that the figure compares the recovery of latent trait θ_j only. This is because 1) there were no latent positions z_j and w_i and the weight parameter γ in the GRM and 2) β_{ik} estimates of the GRM and the data-generating values of β_{ik} used in the LSGRM are not comparable under substantial CD. Regarding the β_{ik} parameters, as the LSGRM has the negative distance effect, its β_{ik} is shifted positively (to have positively larger values) compared to those in the GRM, unless the data imply very little CD and the spike-and-slab prior removes the distance effect. In fact, β_{ik} in the GRM correspond to the overall sizes of $\beta_{ik} - \gamma \cdot d(z_j, w_i)$ in the LSGRM. However, while β_{ik} is constant for all persons, $\beta_{ik} - \gamma \cdot d(z_j, w_i)$ varies across persons. This makes it infeasible to compare the estimates of β_{ik} between the two models.

The results in Figure 1 highlight the advantages of the proposed LSGRM in recovering model parameters. Relative to the GRM, the LSGRM produced smaller biases, albeit with somewhat larger SEs due to increased model complexity, ultimately resulting in lower MSEs. In contrast, the GRM, without model components to account for CD underlying the data, showed larger biases in the latent trait estimates, leading to higher MSEs. Thus, unless there is statistical evidence supporting the CI assumption, parameter estimates obtained from the LSGRM are more favorable than those from the GRM. As will be demonstrated below, both real-world datasets analyzed in our empirical examples provide evidence of CD, further indicating that this cross-recovery finding has practical relevance.

Figure 2 presents the cross recovery results under the CI condition. The figure configurations remain the same as those in Figure 1. Before examining the

result, it should be noted that in the CI condition, the spike-and-slab prior in the LSGRM chose the spike part across all repetitions and all sample size conditions (this accuracy of the model selection feature of the model will be further supported by the second simulation study in the following section). This reduced LSGRM to the GRM, effectively removing the latent space parameters z_j , w_i , and γ .

As a result, it could be anticipated that the recovery results of the two models would be highly similar. The trends in MSE, bias, and SE confirm this. With the exception of bias of β_{ik} , which was generally very small for both models, all summary statistics were nearly identical between the two models across all simulation conditions. These findings indicate that the LSGRM can achieve as reliable parameter recovery as the GRM, whenever the CI assumption holds. Because it is often difficult to determine whether this assumption is satisfied in real-world applications, the LSGRM may represent a safer modeling option in practice.

The two models can also be compared by their predictive accuracy. We performed posterior predictive checking (PPC) and relevant analysis, which showed that the LSGRM can yield better predictions of person-wise mean responses and item-wise proportions of M response options. The results are presented in our [Supplementary Materials](#) (Section B) to save space.

Study 2: Ability of the spike-and-slab prior to detect conditional dependence in ordinal response data

In the second simulation study, we examine the accuracy of the spike-and-slab prior distribution in detecting CD underlying the data. To this end, we simulate synthetic ordinal response data with and without CD.

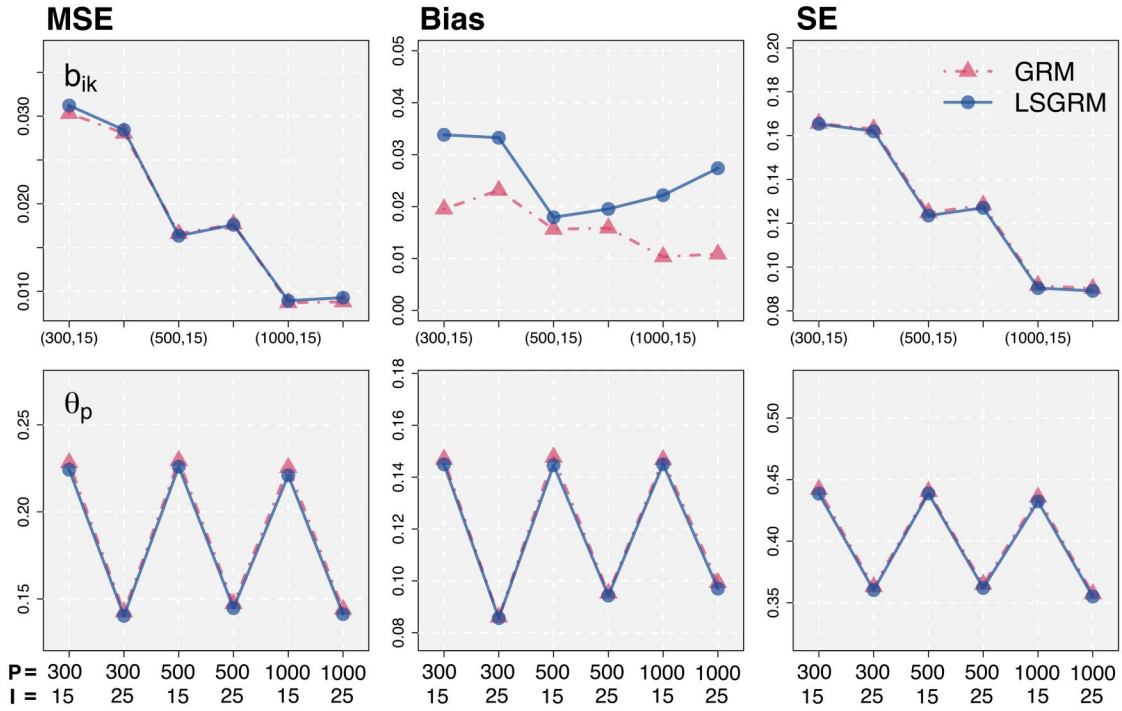


Figure 2. Cross recovery results of the proposed LSGRM and the traditional GRM under the CI condition (i.e., when the data-generating model was the GRM).

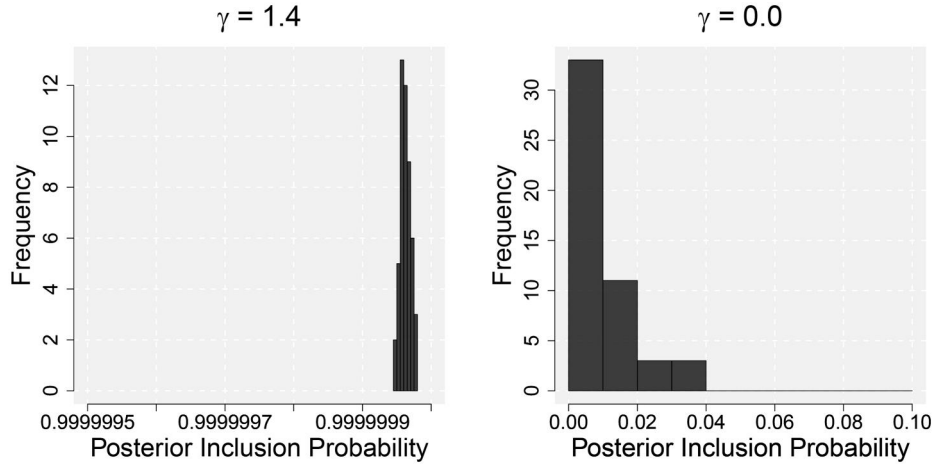


Figure 3. Histograms of the Posterior Inclusion Probability (PIP) across the 100 replications 1) when $\gamma = 1.4$ and there is CD in data (Left) and 2) when $\gamma = 0$ and there is no CD in data (Right).

Motivated by the simulation in Jeon et al. (2021, for the LSIRM), person and item latent positions are assumed to have a specific pattern to induce CD. Namely, there are two separate person clusters (150 persons for each) as well as two item clusters (7 items for each), and each person cluster is closer to one item cluster and farther away from the other. This implies that a group of persons would have a negative dependence with one set of items (unknown a priori) and the other group of persons would have the same trend with the other set of items.

To generate synthetic datasets, we use (a) $\gamma = 1.4$ for the first simulation condition in which CD is

present in the data and (b) $\gamma = 0$ for the second condition in which the data do not imply CD. For each condition, we generate 100 datasets of ordinal responses with $N = 300$ respondents, $I = 14$ items, and $K = 4$ response levels. Then, the proposed LSGRM is fit to datasets with the same estimation method as in the first simulation study.

We compute the PIP values $P(\delta = 1 \mid \gamma)$ per repetition to evaluate the performance of the slab-and-spike approach for both conditions. Figure 3 shows the histogram of PIP values across all repetitions in the simulation conditions with $\gamma = 1.4$ (left) and with $\gamma = 0$ (right). In the first condition with $\gamma = 1.4$, the

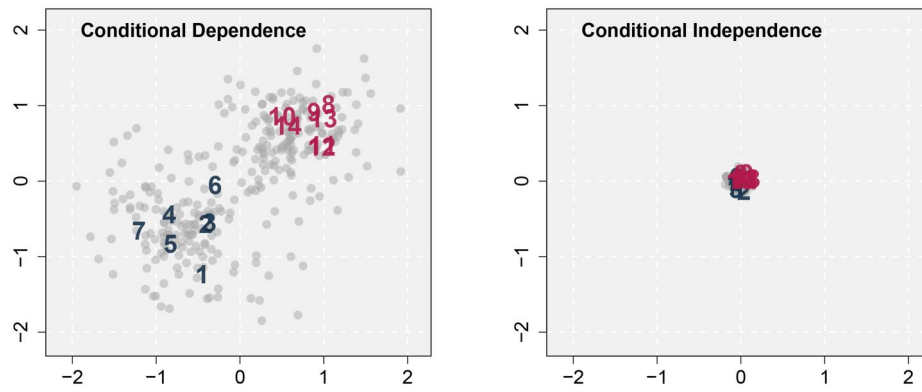


Figure 4. Interaction maps with latent positions of items (numbers) and respondents (points) 1) when $\gamma = 1.4$ and CD is present in data (Left) and 2) when $\gamma = 0$ and there is no CD in data (Right).

data imply CD that cannot be captured by the original GRM parameters, and the spike-and-slab prior in the LSGRM correctly detected this. After burn-in, the computed PIP values were always $> .999$ for all repetitions. In contrast, the proposed model in the second condition yielded small and close-to-zero PIP values for all repetitions. That is, the spike-and-slab prior again successfully rejected the additional parameters to capture CD, reducing the model to the simpler GRM. This demonstrates that the model selection feature of the proposed model can effectively remove the distance effects and reduce to the simple GRM when there is no substantial CD.

We further investigate if the proposed model can correctly recover the assumed structure of latent positions, beyond simply detecting substantial CD. To this end, we obtain the estimated latent positions and recover their interaction map. For illustrative purposes, the first repetition per condition is selected and the corresponding latent space is presented in Figure 4.

Panel (a) of Figure 4 illustrates the results for the condition where the data imply CD ($\gamma = 1.4$). The estimated latent positions of persons (represented as gray dots) and items (numbers) form two distinct clusters. One cluster of persons is positioned closer to the first seven items, while the other cluster is closer to the last seven items. In contrast, Panel (b) shows the no-CD condition ($\gamma = 0$) in which all person and item positions are densely concentrated near the origin of the latent space. As well as estimating γ as near-zero through the spike-and-slab prior, the model also estimated all latent positions around 0 and yielded close-to-zero distances. This corresponds to no interactions assumed in data generation for this condition. Overall, PIPs and recovered interaction maps demonstrate that the proposed model can correctly detect or reject CD and recover the underlying structure of person-item interactions.

Empirical examples

We present two empirical examples to illustrate and compare the proposed LSGRM approach in real-world assessment settings. The first example focuses on illustrating the advantages and utilities of integrating the latent space into the GRM. The second example demonstrates the impact of model misspecification on CD, estimated latent spaces, and resulting person-item interactions.

Example 1: Illustration of the LSGRM with the maternal depression data

The first empirical study aims at illustrating the proposed model in a real-world context, focusing on the interpretation of the interaction map results it produces.

Data description

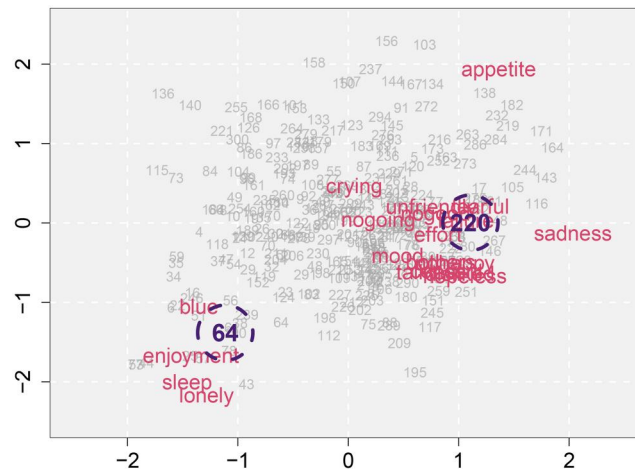
We used data from Santos et al. (2018), who examined perinatal depression symptoms in a cohort of 306 low-income mothers. The mean maternal age was 26.1 (with $SD = 5.8$) and the mothers lived with a mean of 1.8 children with a maximum age of five ($SD = 0.8$) in a household. About 50.5% of the mothers lived without a spouse or partner. Also, 41.3% had a full- or part-time job while 17% were still in school, and the rest (41.6%) had neither of the previous. The ages of the recently-born child in a household were between 6 weeks and 36 months, with a mean of 22 months ($SD = 13.6$), and 51.9% of the children were females. The mean level of education of the mothers was 11 years ($SD = 2.8$).

The depression levels of these mothers were assessed using the Center for Epidemiological Studies Depression Scale (CES-D; Radloff, 1977). This validated self-report tool can screen for 20 symptoms of depression, providing a comprehensive profile of depressive symptoms with a broad spectrum. With the CES-D items, the mothers were asked to report their

Table 2. Mean and Standard Deviation (SD) of depressive symptoms from the clinical sample of low-income mothers in Santos et al. (2018).

Abbrev	Symptom	Mean (SD)	Abbrev	Symptom	Mean (SD)
bother	Feeling bothered	2.31 (0.96)	sleep	Sleep disturbance	2.27 (1.03)
appetite	Appetite changes	2.71 (1.05)	nohappy	Lack of happiness	2.26 (1.05)
blue	Feeling blue	2.25 (1.10)	talkless	Talking less	1.98 (1.03)
nogood	Lack of feeling good	2.55 (1.00)	lonely	Lonely	2.45 (0.96)
concent	Concentration difficulties	2.74 (0.98)	unfriendly	People unfriendly	2.22 (0.89)
mood	Depressed mood	1.87 (0.98)	enjoyment	Lack of enjoyment	2.42 (1.04)
effort	Everything was an effort	2.62 (1.11)	crying	Crying	1.94 (1.01)
hopeless	Hopelessness	2.09 (1.03)	sadness	Sadness	2.21 (1.15)
failure	Feeling of failure	2.11 (1.06)	disliked	Feeling disliked	2.64 (1.06)
fearful	Fearful	2.61 (1.16)	nogoing	Inability to get going	1.97 (1.08)

The Center for Epidemiological Studies Depression Scale (CES-D) was used for assessment.

**Figure 5.** Interaction map of the Maternal Depression data estimated from the LSGRM. Gray numbers represent the latent positions of mothers and words represent depression symptom items abbreviated according to Table 2. Two respondents (with IDs of 64 and 220) are selected for further illustrations. Their latent positions are highlighted with a larger font size, purple color, and dashed circles.

depressive symptom severity during the previous seven days. Each symptom was scored from 1 (not at all) to 4 (5–7 days), related to a statement expressed in each item. Table 2 lists the CES-D symptoms with the mean and SD of item responses from the mothers. Two mothers were excluded because they did not respond to the CES-D, leaving $N = 304$ for our main analysis.

Overall results

We fitted the LSGRM to the depression dataset with the same estimation method as used in the simulation studies. Convergence was assessed with trace plots and $\hat{R}$ diagnostics, supporting reasonable convergence (see Supplementary Materials - Section E). The spike-and-slab prior detected non-ignorable CD unexplained by the person and item parameters (latent traits and item thresholds) with $\text{PIP} > .999$, implying violations of the CI assumption of the traditional GRM. Accordingly, the posterior mean of γ was 1.644 with SD of 0.093 and 95% credible interval (CI) of

[1.472, 1.839]. Therefore, we further explored CD via the estimated latent positions of persons and items.

Figure 5 presents the interaction map of the depression data. In this figure, gray numbers represent the latent positions of mothers and words indicate those corresponding to depression symptoms, abbreviated according to Table 2. From this map, distances between respondents, between symptoms, and between respondents and symptoms can be evaluated. Importantly, respondent-symptom distances can be examined to study respondent-item interactions implied by CD. In this case, larger distances indicate lower likelihoods of the mothers endorsing the symptom more strongly, beyond what can be predicted by the estimated overall symptom level and item thresholds.

A close inspection of the latent space suggests that most symptoms are distributed on the right side of the origin (0,0), forming a central cluster. In addition, a second, smaller cluster is also observed at the far bottom-left side of the space, which includes symptoms such as *blue*, *enjoyment*, *sleep*, and *lonely*.

Additionally, two items, *appetite* and *sadness*, are positioned far apart, with *appetite* located at the top-right and *sadness* slightly to the right, both isolated from the other items. These clusters and isolated symptoms imply dissimilarities of the depression symptoms that are distinct from variations in their item threshold parameters. It is worth noting that, even within the central cluster, symptoms exhibit unique characteristics. For instance, symptom *crying* is positioned slightly at the top-left side of the cluster, although the distance is smaller than those to the other cluster and the isolated items. In fact, each specific symptom has its own distance profiles relative to other symptoms, which can be utilized to shed light on unexplored symptom-specific features.

The distribution of mother latent positions can be analyzed similarly. Most positions are broadly dispersed around the origin, forming a large circular distribution. Notably, there is a group of mothers that is highly overlapped between *nogoing* and *mood*, forming a small but dense (i.e., thicker gray) cluster. This suggests that these mothers would share similarities in how they endorsed items on the questionnaire, after controlling for the heterogeneity captured by their latent traits. Additionally, some mothers are positioned in the bottom-left region of the space, closer to the smaller cluster of items. With relatively little decreasing distance effects, these mothers are anticipated to endorse the symptoms in this cluster more strongly than others, corresponding to what would be predicted based solely on their overall depression levels and item parameters.

We further elucidate the distances described above in the next subsections to illustrate the utilities and advantages we can draw from the estimated latent space. To ease the interpretation of distances, we introduce the exponential decay transformation of the interaction term, $\exp(-\gamma \cdot d)$, where d is the distance computed from the latent space. This transformation, widely used in the clustering literature (e.g., Ng et al., 2001) as well as some previous cognitive/mathematical models in psychology (e.g., scale-independent memory, perception, and learning, SIMPLE model of memory, Brown et al., 2007; Generalized Context Model of Categorization, Nosofsky, 1986) defines similarities. This has advantages over distances in that 1) (unlike distances) it has a monotone and positive relationship with logit response probabilities and 2) it scales distances onto the (0, 1], both allowing a more straightforward interpretation and comparison of distances and interactions. Also, according to Equations 2, we can derive:

$$\begin{aligned} \frac{p_{ijk}}{1 - p_{ijk}} &= \exp(\theta_j + \beta_{ik} - \gamma \cdot d_{ji}) \\ &= \exp(\theta_j + \beta_{ik}) \cdot \exp(-\gamma \cdot d_{ji}) \end{aligned} \quad (5)$$

where $d_{ji} = d(\mathbf{z}_j, \mathbf{w}_i)$. This shows that $\exp(-\gamma \cdot d_{ji})$ represents the changes of odds due to person-item interactions, i.e., a decrease in the scaled distance represents an increase in the probability of having a stronger symptom relative to the probability of having a weaker (or no) symptom, similar to the interpretation of a regression coefficient in a logistic regression.

Insights into persons: Deriving personalized diagnoses of respondents

The latent distances from respondents to symptom items can be further studied to derive personalized diagnoses, evaluations, and feedback. To illustrate, we chose two mothers (with ID 64 and 220), whose latent positions were highlighted with the larger font size, purple color, and dashed circles in Figure 5. These two mothers had similar latent scores of depression, with $\hat{\theta}_{64} = 0.242$ and $\hat{\theta}_{220} = 0.203$ when $\hat{\sigma}_{\theta} = 0.977$. Therefore, nearly identical diagnoses and treatments would be given to these mothers if judgments were made based solely on these scores.

Despite this similarity in overall latent traits, these two mothers are positioned far away from each other in the estimated latent space (Figure 5). The mother with ID 64 can be found on the bottom-left side of the space, located close to the small item cluster containing four symptoms, whereas the other mother with ID 220 is positioned around the center-right, close to the big central symptom cluster. Due to this difference in latent positions, these mothers have largely different distance values to items, yielding noticeable heterogeneity in their similarities to symptoms not accounted for by latent traits and item parameters.

The difference between the two mothers can be better illustrated with the individual profiles in Figure 6. In each panel, the bars represent the mother-symptom similarities obtained by transforming latent distances d_{ji} using the exponential decay function $\exp(-\gamma \cdot d_{ji})$. Also, the bars are color-coded to highlight symptoms with high similarities for each mother, with darker red-dish colors representing stronger symptoms. Also, the items are sorted in decreasing order of their similarities to the mother with ID 64 for both panels, to ease the comparison between the two profiles.

Panel (a) shows that the mother with ID 64 is likely to strongly endorse symptoms such as *blue* and

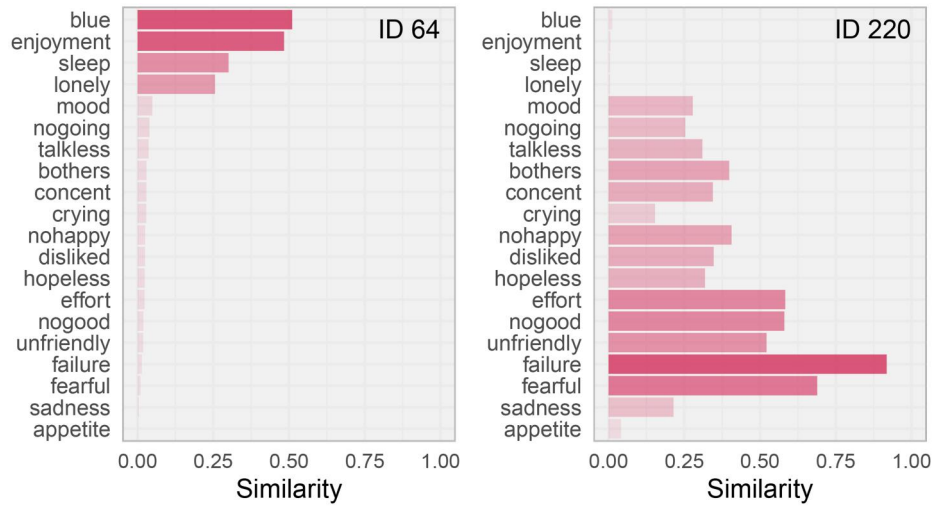


Figure 6. Individual patient profiles based on the interaction map for the two selected mothers (with ID 64 and 220) in the Maternal Depression data. The length of each bar indicates the similarity of the mother and symptom obtained by transforming the mother-symptom distance on the latent space in Figure 5. Higher similarities imply relatively higher likelihoods of endorsement, given latent traits and item parameters. Both mothers have similar overall depression levels ($\theta_{64} = 0.301$ and $\theta_{220} = 0.279$).

enjoyment, and endorse other symptoms such as *sleep* and *lonely* at a medium level, relative to the overall endorsement predicted by the main GRM estimates. Similarities to other symptoms are very low, meaning that these items would be endorsed far weaker than what can be predicted by latent traits and item parameters. In contrast, an individual profile of the other mother with ID 220, displayed in Panel (b), shows that the two main endorsed symptoms are *failure* and *fearful*. Most of the other symptoms in the central cluster of the latent space are expected to be endorsed at the medium-to-high level. However, the items in the small cluster on the bottom-left side of the latent space and one of the isolated item *appetite* have low similarities.

The comparison of the two selected mothers shows the information we can extract from CD and person-item interactions, more than what can be inferred from estimated latent scores and item thresholds. The interaction map can be utilized to examine general patterns of respondents and symptoms that CD implies. Selecting a single or a couple of individuals, this information can be further polished to generate individual profiles of symptoms. This helps us delve into individual characteristics that a respondent exhibits and provide more detailed information on which symptoms are more or less endorsed than the overall judgment made for the respondent. The profile is based on the similarity on the $(0, 1]$ scale, allowing easy evaluations of person-item interactions and comparisons between symptoms as well as between respondents.

Insights into items: Delving into unknown item associations

The distance-based interpretation made in the previous subsection can also be employed to study hidden item associations. For instance, for items i and i' , we can calculate their distance $d_{ii'} = d(\mathbf{w}_i, \mathbf{w}_{i'})$ and then transform it to an item-to-item similarity measure. This was applied to all possible pairs of the depression symptom items and visualized with a heatmap shown in Figure 7.

In this matrix-like heatmap figure, rows and columns represent items (as indicated by the axis labels) and the corresponding cells show the similarity between the row and column items. The cells are color-coded according to the inset legend at the top-left corner, with darker reddish colors representing stronger similarities, whereas more whitish colors indicate weaker similarities. This color-based representation provides a quick overview of item similarities in patient endorsements of depression symptoms and highlights overall item clusters if any. In the figure, the items are ordered based on their latent positions in Figure 5, starting with those on the bottom-left side of the latent space and progressing to those on the top-right.

In general, the heatmap in Figure 7 provides further support for the clustered item structure identified in the estimated latent space (Figure 5). Specifically, the items near the center of the columns, from *crying* to *fearful*, exhibit strong similarities with each other, as reflected by generally pink to red cells in the center

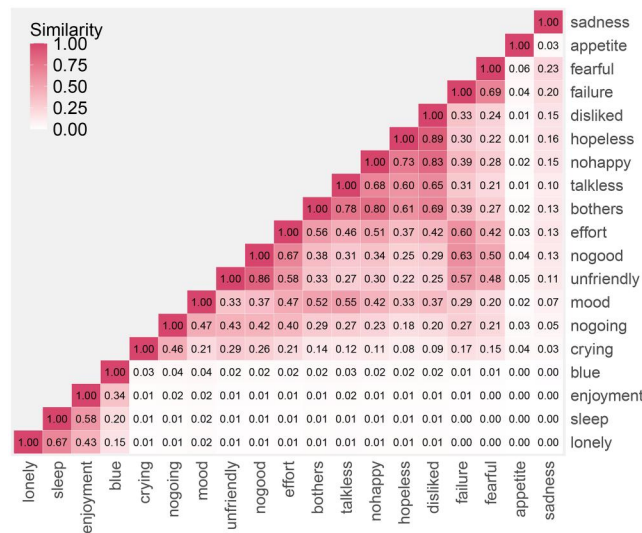


Figure 7. Heatmap of the item similarities from the LSGRM results. The row and column labels represent depression symptoms. The cells represent the similarity between symptom pairs, obtained based on the item distances on the latent space. The cells are color-coded according to the inset legend at the top-left, with more reddish colors indicating stronger similarities whereas more whitish colors indicate weaker similarities in the latent space.

of the item heatmap. These items correspond to the large central cluster in the latent space. In contrast, the first four columns are distinctly separated from this central cluster, as indicated by the white submatrix in the bottom part of the figure. These items belong to the smaller cluster located on the bottom-left side of the latent space, with relatively high intra-cluster similarities. The last two items, *appetite* and *sadness*, are isolated and display weak or no similarity to the other items, consistent with their positions in the latent space.

Example 2: Impact of model misspecification on conditional dependence

In the second study, we examine the impact of misspecifying the type of observed variables on CD and person-item interactions. To this end, we empirically compare the proposed LSGRM against 1) the LSIRM for binary responses and 2) the LSCFA for continuous responses, both fit to the ordinal response data based on the Likert scale. For the LSIRM, we dichotomized ordinal response data and fitted the model. For the LSCFA, the model was fitted to the ordinal response data despite the misspecification. The resulting latent spaces, individual profiles, etc., were compared with those from the LSGRM.

Data description

For this example, we analyzed data from the Internet Addiction Test (IAT), previously studied by Manda et al. (2019) and Mwakilama et al. (2022). This dataset

Table 3. Mean and Standard Deviation (SD) of internet addiction behaviors from the Internet Addiction Test dataset (Manda et al., 2019; Mwakilama et al., 2022).

Abbrev	Mean (SD)	Abbrev	Mean (SD)
online	3.60 (1.26)	anticipate	3.11 (1.40)
chores	2.80 (1.30)	empty	3.70 (1.72)
excite	2.87 (1.55)	annoy	2.81 (1.51)
relation	3.08 (1.44)	latenight	3.16 (1.51)
complain	2.64 (1.39)	preoccupied	2.90 (1.47)
school	2.47 (1.32)	glued	3.42 (1.56)
email	2.91 (1.52)	time	3.16 (1.48)
job	2.33 (1.30)	hide	2.88 (1.62)
defensive	2.70 (1.64)	more	2.99 (1.51)
disturb	3.03 (1.52)	mood	3.04 (1.59)

comprised 984 tertiary education students (345 females and 639 males) randomly sampled from 13 institutions in Malawi, from May to July 2018. A multistage sampling technique was used to ensure diverse representation across regions and academic disciplines, including science (749) and humanities/social sciences (233). The age distribution of the participants ranged from 15 to over 30 years, with the majority (57.1%) falling between 20 and 24 years. The sample primarily consisted of undergraduate students (938) and a smaller group of postgraduate students (45).

The IAT consists of 20 items that assess the impact of internet use on daily routines and psychological well-being: social life, productivity, sleeping patterns, and feelings. Participants rated their internet usage on a six-point Likert scale, with higher scores indicating greater levels of internet addiction behaviors. Response options were as follows: 1 = *Does Not Apply*, 2 = *Rarely*, 3 = *Occasionally*, 4 = *Frequently*, 5 = *Often*, and 6 = *Always*. Table 3 lists the IAT items in

their abbreviations with the mean and SD of the item responses. The full item descriptions can be found in [Supplementary Materials](#) (Section B). Eight students were excluded because they did not respond to more than one-third of the IAT items, leaving $N = 976$ for our main analysis.

We fitted the LSGRM to ordinal responses in the IAT dataset with the same estimation method used previously. Convergence was tested with traceplots and $\hat{R}$ diagnostics, supporting reasonable convergence (see Section E in our [Supplementary Materials](#)). For the LSIRM application, we dichotomized the response data in three different ways (described below). Then, we fitted the LSIRM to the dichotomized data with the same Bayesian method and similar non- or weakly-informative priors as used in Jeon et al. (2021), Kang and Jeon (2024), and Luo et al. (2023). Also, the LSCFA model was fit to the original ordinal response data despite the difference in the variable

type, with the same Bayesian approach as used in Kang and Jeon (2025a). Convergence was assessed with the same approach as the LSGRM (see Section F in our [Supplementary Materials](#)).

In all applications of the LSGRM, the LSIRM with the three dichotomization methods, and the LSCFA model, the PIPs were greater than 0.999, implying violations of the CI assumption and the existence of substantial CD. The posterior estimates of γ with its Bayesian SD and 95% are provided in [Table 4](#), as well as their corresponding posterior density plots in our [Supplementary Materials](#) (Figures S8 and S14-S17).

Comparison of the LSGRM and the LSIRM

We introduced three dichotomization methods, denoted as D1, D2, and D3, to examine the effect of distorting the full range of the Likert scale on CD. The three methods are presented in [Table 5](#). For each dichotomization method, the behaviors considered *present* were coded as 1 and those considered *absent* were coded as 0.

[Figure 8](#) summarizes the effect of dichotomization methods on the observed responses. The four histograms show the distribution of person-wise mean responses. Panel (a) presents the histogram from the original ordinal responses on the six-point Likert scale. The overall mean is 3.009 and the distribution is slightly right-skewed, indicating that there are slightly more students with lower overall IAT scores. Panels (b)-(d) show the histograms of person-wise

Table 4. Posterior mean, standard deviation (SD), and 95% credible interval (CI) of the latent space weight parameter γ obtained from the LSGRM and the LSIRM with different dichotomization scenarios (D1-D3) fit to the Internet Addiction Test (IAT) data.

Model	Mean	SD	95% CI
LSGRM	1.027	0.080	[0.872, 1.181]
LSIRM: D1	0.932	0.097	[0.748, 1.128]
LSIRM: D2	1.082	0.089	[0.909, 1.258]
LSIRM: D3	1.089	0.101	[0.887, 1.281]
LSCFA	0.738	0.054	[0.630, 0.846]

Table 5. Three dichotomization methods considered for the application of the Latent Space Item Response Model (for binary responses; Jeon et al., 2021) to ordinal responses in the Internet Addiction Test data (Manda et al., 2019; Mwakilama et al., 2022).

Methods	Absent (0)	Present (1)	Description
D1: half-and-half	1, 2, 3	4, 5, 6	Divided the ordinal responses into two halves based on the original number of points used in the Likert scale.
D2: rare-to-absent	1, 2	3, 4, 5, 6	Up to rare occurrences of behaviors were considered <i>absent</i>
D3: not-apply-to-absent	1	2, 3, 4, 5, 6	Behaviors were considered <i>absent</i> only when respondents chose <i>does not apply</i> .

Note: Response options of the six-point Likert scale were 1 = *DoesNotApply*, 2 = *Rarely*, 3 = *Occasionally*, 4 = *Frequently*, 5 = *Often*, and 6 = *Always*. The ordinal responses (from 1 to 6) were mapped onto binary responses in the leading row.

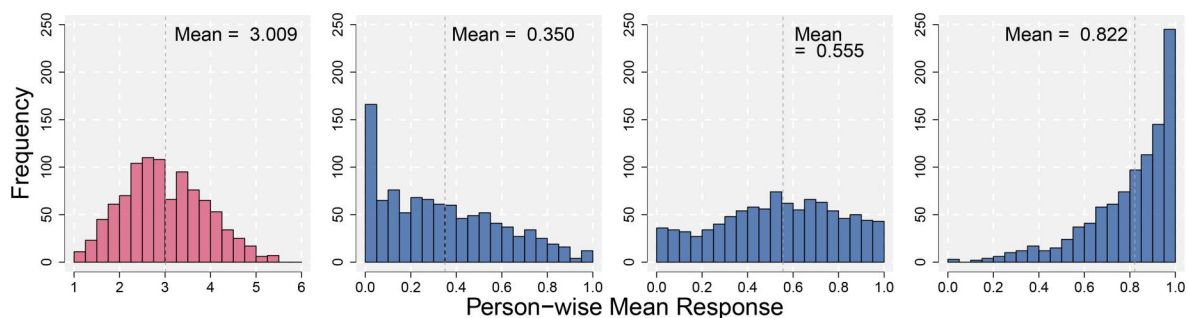


Figure 8. Histograms of person-wise mean responses. Panel (a): Ordinal responses on the six-point Likert scale. Panel (b)-(d): Dichotomized responses with the methods D1-D3, respectively, presented in [Table 5](#).

means of dichotomized responses (i.e., mean proportions). The method D1 (half-and-half) maps ordinal responses $\{1, 2, 3\}$ to 0 and $\{4, 5, 6\}$ to 1. As there were more $\{1, 2, 3\}$ responses in the ordinal data, this mapping produces more 0's, resulting in a right-skewed distribution of person-wise proportions. The method D2 (rare to absent) maps the first two ordinal responses to 0 and the others to 1. The resulting distribution is relatively flat compared to the results of the other two dichotomization methods. The last method D3 (not-apply-to-absent), which maps only the first ordinal response to 0, generated too many 1's after dichotomization, resulting in a heavily left-skewed distribution.

As the spike-and-slab prior confirmed substantial respondent-item interactions (Table 4), we moved on to analyzing the estimated latent space. Below, we focus on illustrating the results on 1) configurations of the estimated latent spaces and 2) individual behavior profiles for a randomly selected illustrative respondent. In doing so, we aim to examine how the effect of dichotomization propagates into the interaction maps and personalized diagnoses derived from the models. We also examined heatmaps of item

similarities for item-level comparisons, but present the corresponding results in our [Supplementary Materials](#) (Section D) to save space.

Estimated Latent Spaces:

Figure 9 displays the interaction maps from the LSGRM (Panel (a)) and the LSIRM with D1-D3 (Panels (b)-(d), respectively). In each panel, the abbreviated words (red or blue) represent the internet addiction behaviors and the grey numbers indicate respondent IDs. To ease visual comparisons, we matched the configurations of different interaction maps by applying Procrustes matching to the LSIRM results, with the posterior mean latent positions from the LSGRM as the reference set.

In the interaction map of the LSGRM, students are widely distributed around the center without notable patterns. Most students are near the origin of the space, with some students distributed on the outer side, constructing a circular-shaped distribution of the latent person positions. Regarding the item positions, most items are distributed around $(w_{1i}, w_{2i}) = (2, 0)$ including *time*, *glued*, and *preoccupied*. Some items might be considered as a separate group, which includes *disturb* and its neighbors slightly above $(1, -1)$.

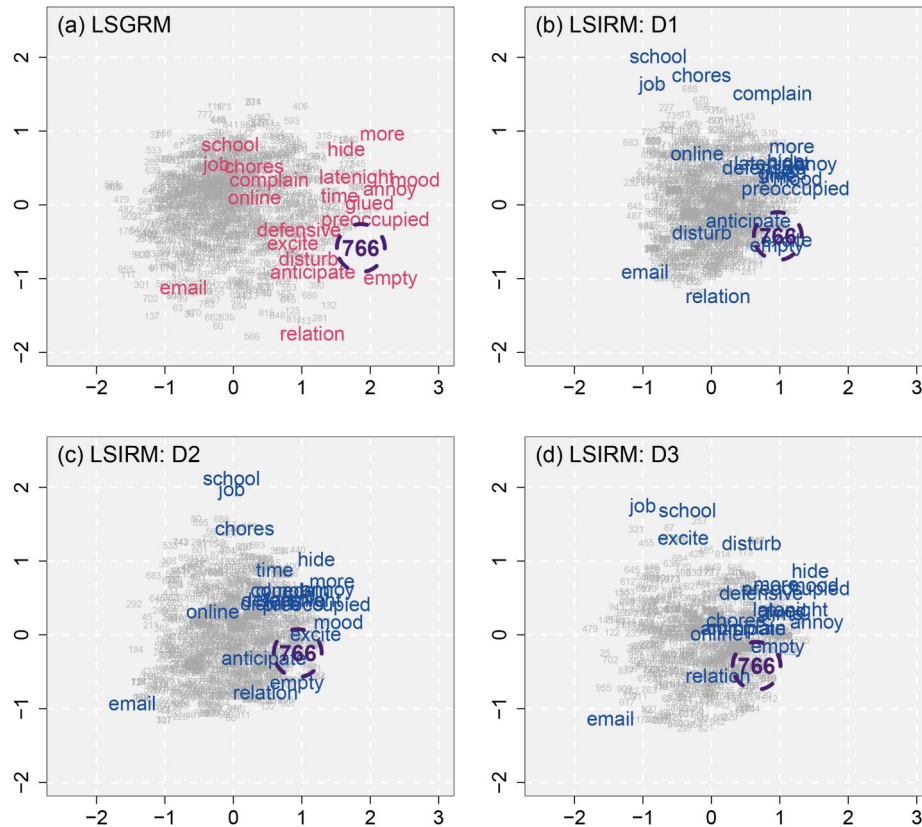


Figure 9. Interaction maps from the LSGRM (Panel (a)) and the LSIRM with the three different dichotomization methods D1-D3 in Table 5 (Panels (b)–(d), respectively). One respondent (with ID of 766) is selected for further illustration. The corresponding latent positions are highlighted with a larger font size, purple color, and dashed circles.

One may identify another cluster of items (although rather close to most of the items) slightly above the origin (with $-1 < w_{1i} < 1$ and $w_{2i} > 0$), which includes the item *school* and its neighbors. The items in the same cluster are anticipated to exhibit similar endorsement patterns by the students, after accounting for differences due to their item effects.

There are also some isolated items such as *email*, and *relation*. These items are expected to have largely distinct patterns of endorsement. Especially, *email* (*How often do you check your e-mail before something else that you need to do?*) is less likely to be endorsed as it is saliently far away from most students. This may imply that frequently checking e-mails does not represent clinical symptoms related to internet addiction, in that an e-mail is not a trigger of internet-addictive behaviors and even highly addicted students do not like to see e-mails.

The general patterns found in the LSGRM result tend to be distorted in the LSIRM results due to dichotomizations. Overall, variability in student latent positions seems to be diminished due to dichotomization. The SDs (across respondents) of the posterior mean student positions were $(SD(z_{1p}), SD(z_{2p})) = (0.704, 0.566)$ in the LSGRM but they were reduced to $(0.484, 0.481)$ for D1, $(0.589, 0.523)$ for D2, $(0.529, 0.465)$ for D3 in the LSIRM. Another related observation is that there are several small gray clouds in the dichotomized versions of the interaction maps. This is because, with dichotomization, more respondents have identical response patterns (therefore, latent positions) more frequently than when ordinal responses are analyzed.

The item positions also showed similar distortions due to dichotomization. This can be seen by comparing item positions across Panels (a)-(d) in Figure 9. We describe some notable differences here. In the D1 scenario, the items in the previously discovered cluster with *school* and its neighbors still form a cluster, though more widely dispersed, on the top side of the latent space. However, item *online* fell away from this cluster, being left around the origin. Items near $(2, 0)$ in the LSGRM results are more densely intermixed in the D1 scenario. This item cluster now includes *defensive*, which was closer to the cluster including *disturb*. In the D2 scenario, a similar merging pattern of items is observed, but item *complain* moved from the cluster of *chores* to the biggest one and the previously isolated *relation* becomes a neighbor of some other items. Lastly, in the D3 scenario, items become closer compared to those in the other scenarios. Some exceptions include the cluster

of *school* and its neighbors, item *disturb* that is now very close to this cluster, and the isolated item *email*.

The item *email* at the bottom-left is the only item that remains consistent across all four results. On average, this item had the lowest correlations with the other items; the mean of the correlations between *email* and the others was 0.196 while those of the other items ranged between 0.194 and 0.549, with mean of 0.338. Also, 85.4% of these correlations were larger than the maximum correlation (0.271) between *email* and the others. It could be deduced that this item-specificity of *email* was captured by its largest distance on the interaction map. For a further inspection of the item similarities, heatmap figures can be found in the [Supplementary Materials](#) (Section D).

Individual Behavior Profiles:

The effect of dichotomization can be examined from a more microscopic perspective, focusing on individuals. To illustrate this, we randomly chose a specific student with ID 766 as an example, and generated the individual profiles of the similarities between this student and internet addiction behaviors, derived from the LSGRM and the LSIRM with the three dichotomization approaches. The ID number of this student is highlighted with the larger font size, purple color, and dashed circle in the estimated latent spaces in Figure 9. The resulting individual student profiles are presented in Figure 10: Panel (a) shows the result from the LSGRM and Panels (b)-(d) display the profiles from the LSIRM with the dichotomization methods D1-D3, respectively. Large spans of colors, from white to red for the LSGRM and from white to blue for the LSIRM were used to help visualize different endorsement patterns (controlling for the effects of the latent trait and item effect parameters) predicted for the same student. A longer bar with a darker color represents a higher level of endorsement whereas a shorter and whiter bar indicates a lower level. To promote comparison across panels, items were sorted in the order of similarities derived from the LSGRM results.

The individual person profiles suggest that using different models and dichotomization methods yields markedly distinct endorsement patterns after the person and item effects are accounted for. Panel (a) in Figure 10 shows that the student with ID 766 would tend to endorse items *preoccupation*, *glued*, and *empty* to the degree almost as close to predicted by the latent trait and the item thresholds, as these items have the smallest distances (and thus high similarities) to the students. In contrast, the students would tend to endorse items *email*, *school*, and *job* much less than predicted by the main model parameters, as these items have the largest distances (low similarities).

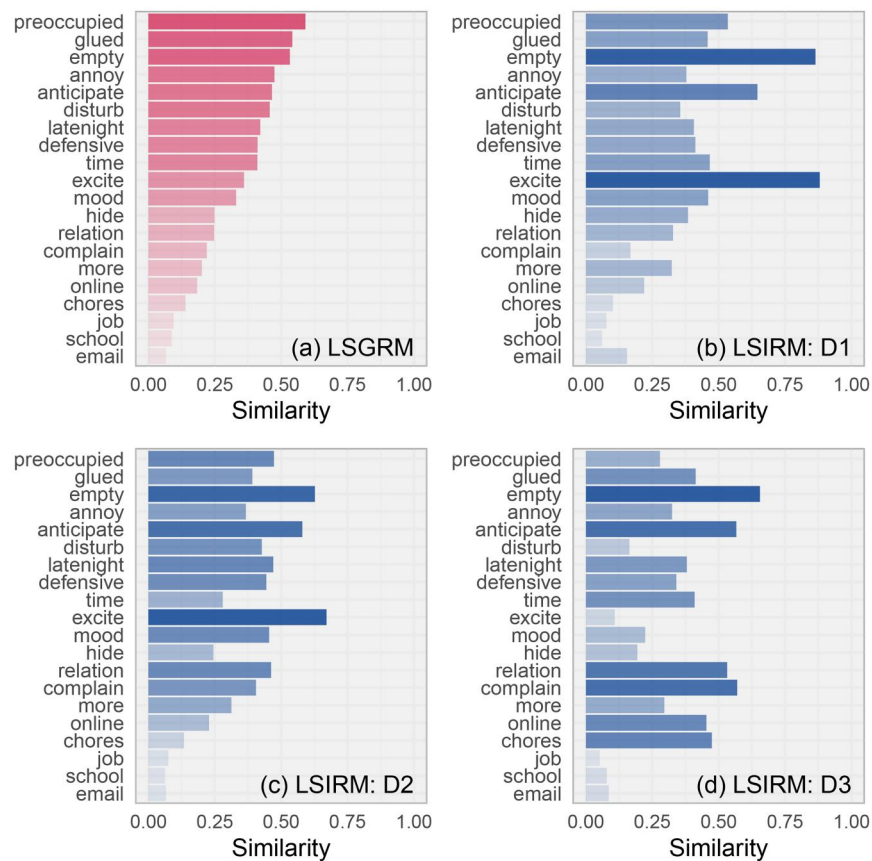


Figure 10. Individual person profiles of symptom similarities of the same student with ID 766, derived from the LSGRM (Panel (a)) and from the LSIRM with three different dichotomization methods D1-D3 in Table 5 (Panels (b)–(d), respectively).

The LSIRM with dichotomization yielded heterogeneity in behavioral patterns of the same student. First, although the three mostly endorsed items (*pre-occupation*, *glued*, and *empty*) still exhibit relatively high similarities, they are not necessarily the top three items in the LSIRM results. Especially, the similarity of *preoccupied* has diminished to a large degree in the D3 scenario whereas it is similar to the LSGRM estimate in the D1 scenario. Also, the item *chores* has nearly uniform similarity estimates across panels (a)–(c) but exhibits a drastic difference in the D3 scenario. Similar differences can be found for most of the items. In particular, similarities of *excite*, *online* and *complain* vary substantially across the four applications.

In summary, the comparison results highlight that individual differences in item response patterns can be considerably altered by modifying responses, such as through dichotomization. Moreover, different dichotomization methods produce varying outcomes of CD, configurations of latent spaces, and individual similarity profiles. Hence, subjective choices of dichotomization can lead to markedly different interpretations of students' endorsement of internet addiction behaviors, ultimately influencing diagnoses and feedback.

Comparison of the LSGRM and the LSCFA

Continuing our investigation, we examined the difference in the estimated latent space and individual profiles between the LSCFA model for continuous responses and the proposed LSGRM. As described before, the spike-and-slab prior implemented in the LSCFA model also yielded the PIP greater than 0.999, implying substantial CD and unexplained person-item interactions as in the LSGRM results.

The top panels in Figure 11 show the estimated latent spaces from the LSGRM (left) and the LSCFA model (right). Note that the top-left panel is the same as Panel (a) in Figure 9 but we repeat the result here for easy comparison. The latent positions derived from the LSCFA model were matched to the reference set from the LSGRM, as done for those from the LSIRM in Figure 9. We made the range of x- and y-axes the same between the two panels so that the latent space from the LSCFA model can be directly compared to that of the LSGRM. Also, small gray numbers represent person latent positions whereas colored words indicate internet addiction symptoms.

The results show a clear reduction in the variability of latent positions in the LSCFA model. Person latent

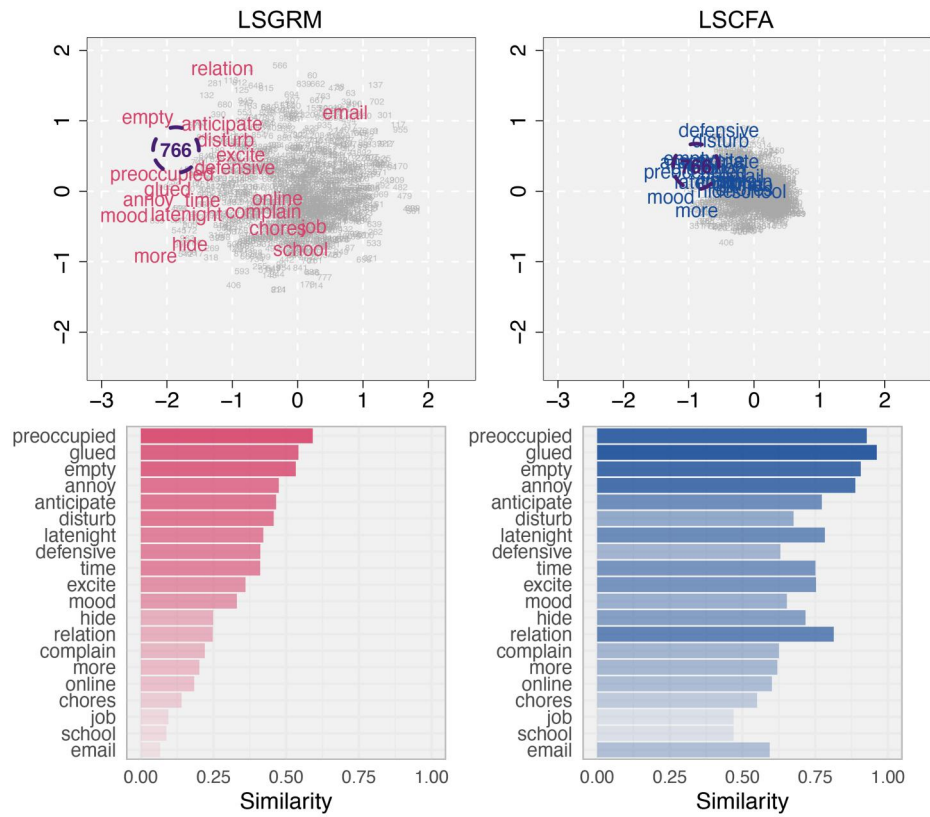


Figure 11. Top: Interaction maps from the LSGRM (top-left) and the LSCFA model (top-right). Bottom: Individual person profiles of symptom similarities of the same student with ID 766, derived from the LSGRM (bottom-left) and the LSCFA model (bottom-right).

positions from the LSCFA model are densely gathered near the origin. The SDs of positions across persons are (0.252, 0.326) for the two dimensions, which are much smaller than those from the LSGRM (0.704, 0.566). Similarly, item latent positions also form a tight cluster near the origin. Some items, such as *defensive* and *disturb*, are located slightly above the cluster, and *mood* and *more* slightly on the left-bottom side, but their distances and configurations do not yield a notable distinction.

Accordingly, individual profiles of similarities to items also exhibit large heterogeneity. Again, we focus on the randomly selected respondent with ID 766, whose latent positions are highlighted with a larger font size, purple color, and dashed circles in the latent spaces in Figure 11. The bottom panels present individual person profiles of this respondent, derived from the LSGRM (bottom-left) and the LSCFA model (bottom-right). As in Figure 10, a longer bar with a darker color indicates a larger similarity. Also, items are sorted in the order of item similarities derived from the LSGRM.

Compared to the profile from the LSGRM, the selected respondent has very small distances to internet addiction behaviors, and accordingly, fairly large similarities. This result indicates that this respondent

would tend to endorse items quite similarly as predicted by the CFA parameters. In other words, although the latent trait, factor loadings, unique variances, and item intercepts were not able to fully describe associations among observed item responses (i.e., the CI assumption was not met), their predictions would be generally close to those incorporating CD.

The observed difference in results between the LSGRM and the LSCFA model may be even more surprising than the LSIRM results presented in the previous subsection, because there was no data transformation (i.e., dichotomization) applied to fit the LSCFA model. Some of the differences could be attributed to model misspecification, as we intended to examine. However, there are other important factors to consider, such as 1) the number of parameters and 2) the types of parameters.

First, the LSGRM may include more or fewer parameters than the LSCFA model. In the present case with $I = 20$ and $M = 6$, the LSGRM has $(M - 1) \times I = 100$ item effect parameters, whereas the LSCFA model has 60 parameters (factor loadings, item unique variances, and item intercepts). With fewer parameters, the LSCFA model would be anticipated to produce larger CD, which does not align

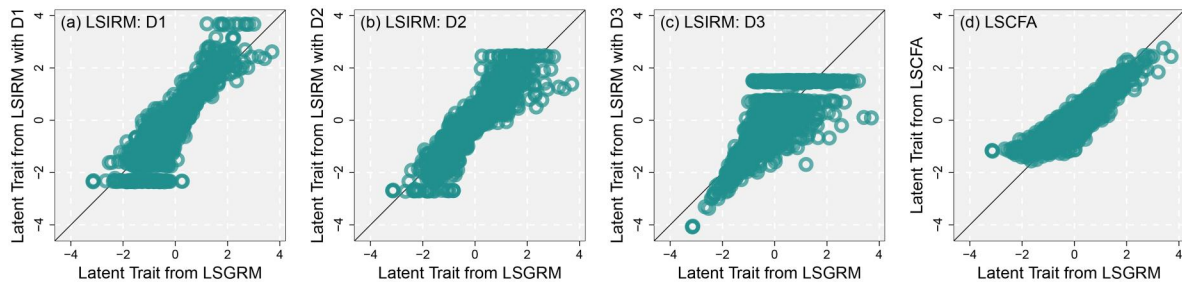


Figure 12. Scatter plots of the estimated latent trait $\hat{\theta}_j$ of the Internet Addiction Test data. Panels (a)–(c) show the comparisons of the LSGRM versus the LSIRM with the dichotomization methods D1–D3, respectively and Panel (d) presents the comparison against the LSCFA model. In each panel, the estimates from the LSGRM are plotted on the x-axis against those from the competing model on the y-axis. The diagonal lines indicate where the estimates are consistent.

with the current finding. Thus, the observed result could be strongly linked to another factor, the type of parameters. Item intercepts in the LSCFA model correspond to the item thresholds in the LSGRM, and may thus account for similar item-specific variations. However, factor loadings and unique variances can represent distinct aspects of item effects (although even they were not sufficient to satisfy the CI assumption) that cannot be explained by the LSGRM. In particular, the product of factor loadings and latent traits in the CFA model represents systematic person-item interactions (Kang & Jeon, 2025b). Such interactions cannot be captured by the main-effect parameters of the current LSGRM but are instead represented through the latent space. It could be the case that these aspects have appeared in the latent space estimated of the LSGRM.

Impact of misspecification on latent traits

It can be expected that applications of the binary LSIRM with dichotomization and the continuous LSCFA model, despite model misspecification, influence not only CD and latent spaces but also the main person effect. Specifically, these inappropriate applications could affect the estimates of the latent traits. To examine this, we compared across the three models the posterior means of the latent trait $\hat{\theta}_j$, representing overall internet addiction levels. For the student with ID 766, it was estimated as $\hat{\theta}_{766} = 1.631$ (93.8% percentile) by the LSGRM. However, it changed to 1.357 (83.5% percentile), 0.549 (70.3% percentile), and -0.201 (42.6% percentile) when estimated by the LSIRM with D1, D2, and D3 dichotomizations, respectively. The LSCFA yielded 1.053 (88.8% percentile).

The same comparison of latent traits can be extended to the whole students in the IAT data. Panels (a)–(c) in Figure 12 present the scatter plots of $\hat{\theta}_j$, comparing the results from the LSGRM versus

those from the LSIRM with the dichotomization methods D1–D3, respectively, and Panel (d) shows the same comparison against the LSCFA model. In each panel, the estimates from the LSGRM are plotted on the x-axis against those from the competing model on the y-axis. The diagonal lines indicate where the estimates are consistent.

All four panels show substantial reductions of variability in $\hat{\theta}_j$ within a specific range. For the LSIRM results, this can be seen by several groups of green dots spanning through seemingly the same horizontal lines. For example, in the bottom-left side of Panel (a), there is a group of individuals with $\hat{\theta}_j$ between about -4 and 0.5 from the LSGRM (x-axis) but the estimates of the same individuals from the LSIRM were nearly the same as -2.25 . As shown in Panel (a) of Figure 8, there were relatively more students with lower overall IAT scores but their item responses were mapped to 0 by D1. This erroneously removed heterogeneity in item responses among these low-IAT students, leading to nearly identical estimates of the latent trait.

A comparable pattern appears in Panel (c), particularly at the top-right side. In the D3 condition, item responses from categories 2 through 6 on the six-point Likert scale were collapsed. This merging eliminated much of the variability among students with moderate to high IAT scores and produced nearly identical latent scores for them. The D2 results also show a similar pattern at both the bottom-left and the top-right sides of the panel, though the reduction in variability there is less pronounced. By contrast, the LSCFA model did not reveal such patterns because no dichotomization was performed. Nevertheless, the estimated latent traits still reflected fewer individual differences, likely because various item parameters, such as factor loadings, unique variances, and item intercepts, already absorb part of the variability present in the data.

Taken together, the comparisons of the individual person profiles for the selected student, alongside estimates of latent traits for both the selected student and the whole sample, highlight the importance of modeling ordinal data without altering responses and with an appropriate link function. Dichotomization diminishes variability in the main person effects by eliminating heterogeneity in ordinal responses across individuals. This reduction leads to residual data variations that the CI models cannot fully account for. Misspecifying the link function also yields misfit, which influences the estimates of main-effect parameters as well as CD. While latent space models can capture some of these residuals, the resulting latent spaces may differ substantially. Employing a model specifically designed for ordinal data helps avoid these unintended effects, enabling more accurate estimation of both latent traits and positions. This, in turn, allows for more adequate evaluations of overall clinical symptoms and personalized diagnoses and assessments.

Discussion

In this paper, we proposed a novel modeling approach for ordinal item responses, building on the one-parameter graded response model. This approach aims to explore unobserved interactions between respondents and items that may go undetected by main effect parameters, such as latent traits and item thresholds. These respondent-by-item interactions reflect CD in the data, which traditional main-effect-only models may fail to account for, making them inadequate for examining CD.

The core idea of the proposed approach is to embed respondents and items within a shared latent space, referred to as an interaction map, and model their interactions based on the geometric distances within this space. This idea is inspired by the LSIRM (Jeon et al., 2021), which integrates the latent space into the standard Rasch model for binary data to test the presence of CD and explore person-item interactions. Our model extends this framework by building on the GRM for ordinal data. This extension is crucial not only for advancing statistical modeling of CD but also for doing it with an appropriate psychometric link between ordinal responses and their latent structure. Furthermore, all these integrative approaches help us identify and leverage CD as well-tailored information to derive customized diagnoses and evaluations for both respondents and items.

Through simulation and empirical studies, we demonstrated that the proposed approach is an effective tool for examining respondent-by-item interactions implied by CD in ordinal item response data. The first simulation study showed that the LSGRM can adequately recover its parameters. Following this, the second simulation study demonstrated that the spike-and-slab prior implemented in the proposed model can correctly detect CD underlying data as well as reject it when the main effect parameters are sufficient to account for response patterns. The resulting latent spaces also appropriately reflect the assumed structure of latent person and item positions.

With the first empirical application, we demonstrated the utility of the proposed approach for visualizing and quantifying respondent-item interactions. This approach can be further adapted to generate personalized symptom profiles and to capture item-to-item associations. Information on CD represented in the latent space allows the calculation of respondent-item distances, which can be processed into individual similarity profiles, as shown in Figure 6. Combined with estimated latent scores, these profiles could be crucial in clinical contexts to develop customized diagnoses and treatment plans for patients. Whereas sum scores and general latent scores from traditional psychometric models provide only an overall assessment of patients, profiles derived from the latent space offer a more sophisticated account of hidden individual differences. By leveraging these profiles, clinicians can target more strongly endorsed symptoms to design more effective treatments. Moreover, patients with similar symptom spectrums can be identified, facilitating the examination of their shared characteristics and the underlying nature of clinical symptoms.

The same distance approach can also be extended to calculate item-to-item distances and corresponding similarity measures. This approach was showcased in Figure 7 as an item heatmap. Beyond simply confirming the general patterns of item associations that emerged in the latent space, this heatmap allows a more focused exploration of the symptom network. Clinicians, or more broadly psychologists, can utilize this visualization to explore which symptoms within the same disease are interdependent and which ones are isolated, more than what can be explained by the main parameters in the psychometric models. Leveraging these newly identified dependencies, more effective therapies and treatments could be developed by relieving similar symptoms together and/or pinpointing isolated symptoms.

The second empirical demonstration regarding the consequences of 1) dichotomization and 2) treating ordinal scales as continuous could be of particular importance for modeling person-item interactions in practice. It adds to the previous studies showing that both kinds of model misspecifications can yield biased parameter estimates and inaccurate statistical decision-making, while also compromising the interpretability of results (e.g., MacCallum et al., 2002; Russell & Bobko, 1992; Verhulst & Neale, 2021). In particular, biases in estimates of psychometric models imply inaccurate representations of main person and item effects, leading to potential distortions of CD. Our findings showed that both misspecifications of ordinal responses with dichotomization or treating them as continuous can largely shrink factor scores and distort individual profiles extracted from a latent space, which leaves potential risks of misguided inferences. In this sense, the development of the LSGRM can be a useful addition to the latent space approaches in psychometrics, offering a suitable choice for ordinal responses (with accounting for CD), just as the GRM has served for the previous IRT models.

The LSGRM and other integrations of latent space and psychometric models offer fundamentally different and more general ways of exploring CD. Unlike statistical tests for differential item functioning (Magis et al., 2010) and measurement (in)variance (Meredith, 1993; Molenaar, 2021), which evaluate CD between pre-specified groups, the proposed modeling approach allows for studying CD between individuals and person-item interactions. The LSGRM also has advantages over other psychometric models designed to accommodate certain forms of CD. For instance, the testlet IRT models (Jiao et al., 2012; Wang & Wilson, 2005) can explain CD among items under the same testlet. The bi-factor models (Liu & Thissen, 2012) can capture CD within groups of items by introducing their corresponding specific factors. The two-tier models (Cai, 2010) further extend the bi-factor structure through an additional hierarchical layer of latent variables to represent CD. However, all these models require researchers to explicitly specify the structure of CD in advance, which is feasible only in restricted situations. In contrast, the LSIRM and LSGRM can discover unknown dependencies between items, between persons, and between persons and items without assuming a pre-defined structure. Moreover, even after applying testlet, bifactor, or two-tier models, residual person-by-item dependencies may remain. This type of CD can be captured by the LSGRM, the LSIRM, and other latent space measurement models

that can incorporate appropriate multidimensional structures (Kang & Jeon, 2025a, b).

Despite this potential advantage, the current study has its limitations and opens several opportunities for future research. First, the scope of the simulation study for validation was somewhat constrained. To generalize the findings further, it would be necessary to expand the simulation scenarios, considering a broader range of data-generating conditions such as uniform or skewed distributions of latent traits. Also, the comparison against the LSIRM (using different versions of dichotomized data) and the continuous LSCFA (applied to ordinal data) was conducted solely on empirical data. Additional validations through simulated data would enhance the robustness of the findings. We leave these and other potential investigations for future research.

The current approach relies on a two-dimensional latent space and the Euclidean distance, but these choices can also be questioned. Although they follow earlier studies in latent space modeling (e.g., Handcock et al., 2007; Hoff et al., 2002; Jeon et al., 2021; Kang et al., 2023; Smith et al., 2019), bivariate latent positions were motivated solely by ease of visualization rather than by an optimal account for CD and unobserved person-item interactions. Future work may tackle this issue by examining whether model selection criteria can identify the optimal latent space dimensionality. A complication is that CD, even when present to a substantial degree, represents residual dependence that takes a relatively smaller portion of the total data information (Kang & Jeon, 2025b). Indeed, latent traits and item effect parameters can still explain large proportions of data variations even when they fail to meet the CI assumption. In contrast, most model selection criteria evaluate overall model fits rather than providing fits specific to CD. This discrepancy creates a fundamental challenge for existing selection criteria in determining the appropriate dimensionality of the latent space.

The Euclidean distance has also been widely used as it is one of the most straightforward measures on the real-numbered latent space. However, this choice is not necessary. For example, the distance measures such as city-block distance and maximum distance (as discussed in Jeon et al., 2021) can also be utilized depending on research purposes. Even further, non-distance measures such as the inner product or projection of latent positions can be employed as in the projection model for network data analysis (Hoff et al., 2002). Advantages and disadvantages of these

different options, compared to the current Euclidean distances, remain to be studied.

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